A3-PS015-TC001

Explainable Home and Last-Mile Connectivity Diagnosis

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Problem Statement & Limitations

Unloaded Speed Tests

Measure peak download bandwidth while missing packet loss, round-trip latency, and loaded jitter.

Diverse Access Media

Performance fluctuates widely across fiber, cellular, fixed wireless, and satellite connections.

Lack of Transparency

Traditional tools fail to explain findings with evidence or generate clean diagnostic escalation reports.

Core Foundation: RFC 2330

Active Probing Metrics

Standardized methodologies for measuring packet loss, round-trip delay, and availability.

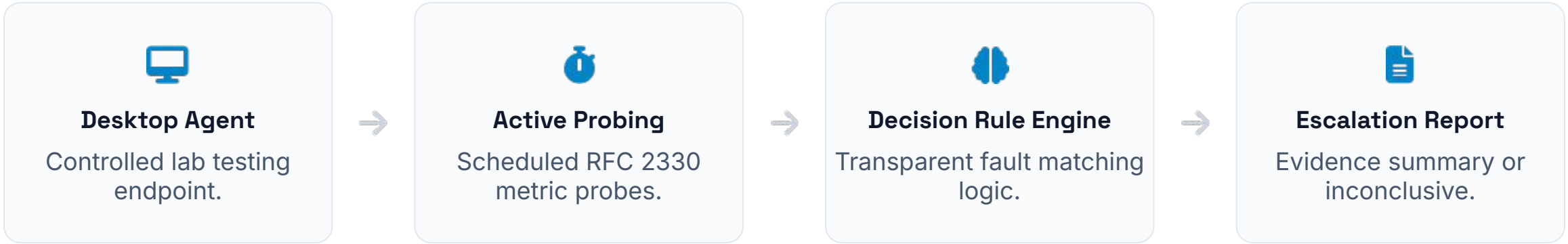
Path & Cloud Definitions

Mathematical framework defining packet behavior across intermediate hops and endpoints.

Error & Clock Accuracy

Rigorous guidelines accounting for clock drift and synchronization errors during measurements.

Proposed Solution Architecture



Testing the 4 Controlled Faults

Gateway Unreachable

Local ARP & ICMP gateway check.

DNS Failure

Targeted UDP/TCP name resolution probe.

Induced Packet Loss

Quantifying drop % and jitter variance.

Network Congestion

Loaded vs unloaded RTT delta tracking.

Key Requirements & Scope Boundary

Core Responsibilities

- Transparent decision rules for diagnosis.
- Handling inconclusive states cleanly.
- Evidence-based reporting.

Out of Scope

- No automated network remediation.
- No definitive damaged-cable detection.
- No ISP-internal root cause claims.

Project Execution Timeline

Phase 1

Setup & Lab

GitHub repository initialization and configuration of the controlled lab testbed.

Phase 2

Fault Injection

Scripting 4 controlled faults and implementing RFC 2330 matching rules.

Phase 3

Validation & Demo

Inconclusive scenario analysis, final technical demonstration, and reporting.

Bibliography & Standards

[1] RFC 2330:

V. Paxson, G. Almes, J. Mahdavi, and M. Mathis, "Framework for IP Performance Metrics", RFC 2330, Internet Engineering Task Force (IETF), May 1998.

[2] ITU-T Y.1541 :

International Telecommunication Union, "Overall network performance objectives for IP-based services", ITU-T Recommendation Y.1541, 2011.